EE4-07 Coding Theory

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. (The Elementary)

(a) Let $\mathcal{C} \subset \mathbb{F}_2^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_2^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_2^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over an binary symmetric channel. Let the received word be $\boldsymbol{y}_1 = [0\ 0\ 1\ 1\ 1\ 1\ 0]$. Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$, and the corresponding transmitted message $\hat{\boldsymbol{m}}_1$. [3]

iv Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over an binary erasure channel. Let the received word be $\boldsymbol{y}_2 = [1\ 0\ 1\ 1\ ?\ ?\ 0]$. Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$ and the message $\boldsymbol{m}_2$. [3]

(b) Let $\mathcal{C}$ be a linear code. Its distance can be defined as

$$d(\mathcal{C}) = \min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\boldsymbol{0}} \mathrm{wt}(\boldsymbol{c}).$$

i State the relationship between the distance $d(\mathcal{C})$ and the columns of its parity check matrix $\boldsymbol{H}$. Prove your answer. [3]

ii What is the distance of the binary linear code defined in Question 1.(a)? Justify your answer. [2]

(c) i Let $a(x) = x^2 + 1 \in \mathbb{F}_2[x]$ and $b(x) = x^3 + x^2 + 1 \in \mathbb{F}_2[x]$. Find $f(x) = \gcd(a(x), b(x))$. [2]

ii Apply Bï¿œzout's identity to compute $g(x) \in \mathbb{F}_2[x]$ and $h(x) \in \mathbb{F}_2[x]$ such that $f(x) = g(x)a(x) + h(x)b(x)$. [2]

(d) Consider the finite field $\mathbb{F}_{16}$ detailed below

	$\mathbb{F}_2[z]/z^4+z+1$		$\mathbb{F}_2[z]/z^4+z+1$
0	0	z^7	z^3+z+1
1	1	z^8	z^2+1
z	z	z^9	z^3+z
z^2	z^2	z^{10}	z^2+z+1
z^3	z^3	z^{11}	z^3+z^2+z
z^4	$z+1$	z^{12}	z^3+z^2+z+1
z^5	z^2+z	z^{13}	z^3+z^2+1
z^6	z^3+z^2	z^{14}	z^3+1

i Compute $\beta_1 = (z+1)\cdot(z^3+z^2+z+1) \in \mathbb{F}_{16}$. [2]

ii Compute $\beta_2 = (z^3+z^2+1)^{-1}\cdot(z^2+1) \in \mathbb{F}_{16}$. [2]

iii Compute the polynomial $f(x) = (x-(z^2+z))\cdot(x-(z^2+z+1)) \in \mathbb{F}_{16}[x]$. [2]

2. (Prime Numbers)

(a) i Let p be a prime number, $a > 1$ and $b > 1$ be two integers. Prove that $p|(ab)$ implies that p divides one of the numbers a and b, or equivalently, if $p \nmid a$ then $p|b$. [3]

ii Let p be a prime number, and $a_i > 1$, $i = 1, \cdots, n$, be integers. Use mathematical induction to prove the statement that $p|(a_1 a_2 \cdots a_n)$ implies that $p|a_i$ for some $i \in \{1, 2, \cdots, n\}$. [3]

iii Find an example where $a > 1,\ b > 1,\ c > 1$ are integers, $c|(ab)$ but $c \nmid a$ and $c \nmid b$. [2]

(b) Let $a, b \in \mathbb{Z}^+$.

i Show that if $\gcd(a, b) = 1$, then the multiplicative inverse $a^{-1} \pmod{b}$ exists. [3]

ii Prove that if the multiplicative inverse $a^{-1} \pmod{b}$ exists, then $\gcd(a, b) = 1$. [3]

Hence, it can be concluded that the multiplicative inverse $a^{-1} \pmod{b}$ exists if and only if $\gcd(a, b) = 1$.

(c) Let p be a large prime number. Let g be a primitive element in the finite field $\mathbb{F}_p$.

i What is the computational complexity to compute $g^a \in \mathbb{F}_p$ in general, where $a \in [1, p-1]$ is a large integer? Explain the typical computation method to achieve this computational complexity. [3]

ii What is the computational complexity of compute the discrete logarithmic function $\log_g(b) \in [1, p-1]$ for a given $b \in \mathbb{F}_p^*$? Explain the typical computation method. [2]

iii Consider the following key exchange problem. Alice and Bob want to share a secret key but their information exchange could be observed by their adversary Eve. The goal for Alice and Bob is to share a key without making it known to Eve.
One solution is the famous Diffie-Hellman key exchange scheme. Suppose that Alice picks a secrete integer a and Bob's secrete integer is b. Describe the rest steps of Diffie-Hellman key exchange scheme. [3]

iv Consider ElGamal public key cryptography scheme. Let $a' \in [1, p-1]$ is the private key for Alice. How would Alice construct her public key?

How would Bob encrypt his message m using Alice's public key? And how can Alice decrypt the message from Bob? [3]

3. (Codes)

(a) i Consider a code $\mathcal{C} \in \mathbb{F}_q^n$ where n is the codeword length. Let $M = |\mathcal{C}|$ be the number of the codewords and d be its distance. Hamming (or sphere-packing) bound provides an upper bound on the number of codewords M. State Hamming bound and prove it. [4]
A code that attains Hamming is called a perfect code.

ii Consider a *linear* code $\mathcal{C} \in \mathbb{F}_q^n$ with parameters $[n, k, d]$. State the corresponding Singleton bound and prove it. [2]
A linear code that attains Singleton bound is called a maximum distance separable (MDS) code.

(b) i Let $r > 2$. A binary Hamming code $\mathcal{H}_r$ is defined as the code with a parity check matrix $\boldsymbol{H}_r$ whose columns consist all the nonzero column vectors of $\mathbb{F}_2^r$. Specify the parameters $[n, k, d]$ of the binary Hamming code $\mathcal{H}_r$. [3]

ii State whether a binary Hamming code $\mathcal{H}_r$ is a perfect code or not. Prove your answer. [3]

iii State whether a binary Hamming code $\mathcal{H}_r$ is an MDS code or not. Prove your answer. [3]

(c) Define the generating function of a codeword $\boldsymbol{c} = [c_0, \cdots, c_{n-1}] \in \mathbb{F}_q^n$ as $c(x) = \sum_{i=0}^{n-1} c_i x^i \in \mathbb{F}_q[x]$. A *linear* code $\mathcal{C}$ is called a cyclic code if for all $c(x) \in \mathcal{C}$, it holds that $u(x)\, c(x) \mod x^n - 1 \in \mathcal{C}$ for an arbitrary $u(x) \in \mathbb{F}_q[x]$. For a cyclic code $\mathcal{C}$, choose its generator polynomial $g(x) \in \mathcal{C}$ as the nonzero monic polynomial of least degree.

i Prove that for all $c(x) \in \mathcal{C}$, it holds that $c(x) = u(x)g(x)$ for some $u(x)$. [2]

ii Construct a BCH code on $\mathbb{F}_2$ with $n = 31$ and $d \geq 7$ by finding an appropriate generator polynomial $g(x)$ in the following way.
Find the corresponding cyclotomic cosets C_i, $i = 0, 1, \cdots, 8$. [3]
Let α be a primitive element in $\mathbb{F}_{32}$. Find a generator polynomial $g(x)$ to guarantee $d \geq 7$. Write it using minimal polynomials. [2]

iii Let α be a primitive element of $\mathbb{F}_q$. Construct a cyclic code $\mathcal{C}[n, k] \in \mathbb{F}_q^n$ where $n = q - 1$ by using the generator polynomial

$$g(x) = (x - \alpha)(x - \alpha^2) \cdots (x - \alpha^{n-k}).$$

What can you say about its distance? Justify your answer. [3]

Solution of Question 1.

(a) i

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

[2]

ii

$$\boldsymbol{G} = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

iii The syndrome vector is given by

$$\boldsymbol{s}_1 = \boldsymbol{y}_1 \boldsymbol{H}^T = [0\ 1\ 1]\,,$$

hence

$$\hat{\boldsymbol{c}}_1 = [0\ 0\ 1\ 0\ 1\ 1\ 0]\,,$$

and

$$\hat{\boldsymbol{m}}_1 = [0\ 1\ 1\ 0]\,.$$

[3]

iv The syndrome vector is given by

$$[1\ 0\ 1\ 1\ 0\ 0\ 0]\,\boldsymbol{H}^T = [1\ 1\ 0]\,.$$

Let c_4 and c_5 be the missing symbols in $\boldsymbol{c}$. Then one has

$$[c_4\ c_5] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = [1\ 1\ 0]\,.$$

It is clear that $[c_4\ c_5] = [1\ 0]$. Hence $\boldsymbol{c}_2 = [1\ 0\ 1\ 1\ 0\ 1\ 0]$ and $\boldsymbol{m}_2 = [1\ 0\ 1\ 0]$. [3]

(b) i The distance of a linear code is the minimum number of linearly dependent columns of its parity matrix $\boldsymbol{H}$.
A concise proof is as follows. Any codeword $\boldsymbol{c} = [c_1, c_2, \cdots, c_n] \in \mathcal{C}$

satisfies

$$\sum_i c_i \boldsymbol{h}_i = \mathbf{0},$$

where $\boldsymbol{h}_i$ denotes the i-th column of the parity-check matrix. It is clear that $\mathrm{wt}(\boldsymbol{c})$ gives a number of linearly independent columns of $\boldsymbol{H}$. Hence, $\min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\mathbf{0}} \mathrm{wt}(\boldsymbol{c}) \geq$ minimum number of linearly dependent columns of $\boldsymbol{H}$. On the other hand, any set of linearly dependent columns $\boldsymbol{h}_i$ defines a codeword $\boldsymbol{c} \in \mathcal{C}$. Hence $\min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\mathbf{0}} \mathrm{wt}(\boldsymbol{c}) =$ minimum number of linearly dependent columns of $\boldsymbol{H}$. [3]

ii The distance of the code in Question 1.(a) is 3. Every column is different. Hence every pair of columns are linearly independent and $d(\mathcal{C}) > 2$. There exist three columns that are linearly dependent, for example, columns 1,2,6. Hence the distance is 3. [2]

(c) i

$$x^3 + x^2 + 1 = (x+1)(x^2+1) + x$$
$$x^2 + 1 = x \cdot x + 1.$$

Hence, $f(x) = 1$. [2]

ii

$$\begin{aligned} 1 &= (x^2+1) + x \cdot x \\ &= (x^2+1) + x(x^3 + x^2 + 1 + (x+1)(x^2+1)) \\ &= (x^2+1) + x(x^3 + x^2 + 1) + (x^2 + x)(x^2+1) \\ &= x(x^3 + x^2 + 1) + (x^2 + x + 1)(x^2+1). \end{aligned}$$

Hence, $g(x) = x^2 + x + 1$ and $h(x) = x^3 + x^2 + 1$. [2]

(d) i

$$(z+1)(z^3 + z^2 + z + 1) = z^4 z^{12} = z^1 6 = z.$$

[2]

ii

$$(z^3+z^2+1)^{-1}(z^2+1) = z^{-13}z^8 = z^{-5}$$
$$= z^{10} = z^2+z+1.$$

[2]

Comments

The computations can be highly simplified by using the provided table and the fact that $z^{15}=1$. Some students failed to recoganise that $z^{15}=1$.

iii

$$\begin{aligned}
&(x-(z^2+z))(x-(z^2+z+1))\\
&=(x-z^5)(x-z^{10})\\
&=x^2-(z^5+z^{10})x+z^{15}\\
&=x^2-(z^2+z+z^2+z+1)x+1\\
&=x^2+x+1.
\end{aligned}$$

[2]

Comments

The key point of this question is that we are looking for a polynomial in x, where the coefficients are polynomials in z. This question is designed in such a way that the end result has a very simple form.

Solutions of Question 2.

(a) i Since $p \nmid a$, $\gcd(p, a) = 1$. By Bï¿œzout's Identity, there exist $x, y \in \mathbb{Z}$ s.t. $xp + ya = 1$. Multiply both sides by b, $xpb + yab = b$. Since $p \mid xpb$ and $p \mid ab$, it holds that $p \mid xpb + yab$, i.e., $p \mid b$. [3]

ii The statement is clearly true for $n = 2$ based on the previous sub-question. Suppose that the statement holds for $n = N$ for a given integer $N > 2$. We are going to show that this statement is true for $n = N + 1$.

Let $a = a_1 a_2 \cdots a_N$. Then $p|(aa_{N+1})$. From the previous sub-question, p divides at least one of a and a_{N+1}. Suppose that $p|a$. Then by the hypothesis, $p|a_i$ for some $i \in \{1, 2, \cdots, N\}$. The claimed statement is true. In the other case where $p|a_{N+1}$, the claimed statement also holds. [3]

iii There are infinitely many possible such examples. One of them is as follows. $6 \nmid 2$ and $6 \nmid 3$ but $6|(2 \cdot 3)$. [2]

(b) i As $\gcd(a, b) = 1$, Bï¿œzout's Identity implies that there exist integers x and y such that $1 = ax + by$. Take the modulo arithmetic for both sides. Then $1 = ax \pmod b$. Hence $x = a^{-1} \pmod b$. [3]

ii This can be proved by contradiction. Suppose that $d = \gcd(a, b) > 1$. From the assumption that $a^{-1} \pmod b$ exists, $1 = ax + by$ some integers x, y. As $d = \gcd(a, b)$, it holds that $d|a$, $d|b$, and hence $d|ax + by$. This leads to $d|1$ which contradicts with the hypothesis $d > 1$. [3]

(c) i The complexity of computing the exponential function $g^a \pmod p$ can be as low as $O(\log p)$. Suppose that the binary representation of p has at most $n + 1$ binary digits. Write a into its binary representation $a = d_n d_{n-1} \cdots d_0$ where $d_i \in \{0, 1\}$. Then $g^a \pmod p = (d_n g^{2^n}) \cdot (d_{n-1} g^{2^{n-1}}) \cdots (d_0 g^0) \pmod p$. One can compute $g^2, g^3, \cdots, g^{2^{n-1}}, g^{2^n} \pmod p$ offline and then $g^a \pmod p$ online with complexity $O(\log p)$. [3]

ii The typical way to compute the discrete logarithmic function $\log_g(b)$ is via exhaustive search. This is because the output of $\log_g(b)$ looks random. This is no existing algorithm that computes it efficiently. The computational complexity is $O(p)$ which is infeasible for large p. [2]

iii Let p be a large prime number and g be a primitive element of $\mathbb{F}_p$. The values of p and g are available in public. Alice computes $A = g^a \in \mathbb{F}_p$ and Bob computes $B = g^b \in \mathbb{F}_p$. They exchange these two values. Then

Alice computes $A' = B^a \in \mathbb{F}_p$ and Bob computes $B' = A^b \in \mathbb{F}_p$. It is straightforward to see that $A' = B'$ can serve as the shared secret key. [3]

iv The public key of Alice can be defined as $a = g^{a'} \pmod{p}$.
Bob randomly choose $t \in \mathbb{N}$ and compute $(b^t, ma^t) \in \mathbb{F}_p^* \times \mathbb{F}_p^*$.
To decode the message from Bob, Alice computes $(b^t)^{-a'} \cdot (ma^t) = m \in \mathbb{F}_p^*$. [3]

Solutions of Question 3.

(a) i Let $t = \lfloor \frac{d-1}{2} \rfloor$. Hamming bound states that

$$M \leq \frac{q^n}{\sum_{i=0}^{t} \binom{n}{i}(q-1)^i}.$$

The proof is sketched as follows.
Let $\mathcal{C} = \{\boldsymbol{c}_1, \cdots, \boldsymbol{c}_M\}$. It is straightforward to see that the spheres $S(\boldsymbol{c}_m, t)$ are disjoint. As

$$\bigcup_{m=1}^{M} B(\boldsymbol{c}_m, t) \subseteq \mathbb{F}_q^n,$$

it holds that

$$\sum_{m=1}^{M} \mathrm{vol}\left(B(\boldsymbol{c}_m, t)\right) = M\left(\sum_{i=0}^{t} \binom{n}{i}(q-1)^i\right) \leq q^n.$$

Hamming bound is therefore proved. [4]

ii Singleton bound states $d \leq n - k + 1$. Note for a linear code $\mathcal{C}[n, k]$, its parity check matrix is of the size $(n-k) \times n$. The distance d is the minimum number of linearly dependent columns of $\boldsymbol{H}$. But every subset of $n-k+1$ columns of $\boldsymbol{H}$ is linearly dependent. Hence $d \leq n-k+1$. [2]

(b) i The number of all nonzero column vectors in $\mathbb{F}_2^r$ is $2^r - 1$. The parity check matrix is hence of size $(n-k) \times n = r \times (2^r - 1)$. Hence $n = 2^r - 1$ and $k = 2^r - r - 1$. The distance of a binary Hamming code is clearly 3. Hence a binary Hamming code $\mathcal{H}_r$ has parameters $[2^r - 1, 2^r - r - 1, 3]$. [3]

ii For binary Hamming code $\mathcal{H}_r[2^r - 1, 2^r - r - 1, 3]$, $t = 1$. Hence $\mathrm{vol}(B(\boldsymbol{c}, 1)) = 1 + n = 2^r$. Therefore,

$$\frac{q^n}{\sum_{i=0}^{t} \binom{n}{i}(q-1)^i} = \frac{2^{2^r-1}}{2^r} = 2^{2^r-r-1} = 2^k.$$

Hence, a binary Hamming code is a perfect code. [3]

iii For a binary Hamming code $\mathcal{H}_r$, $n - k + 1 = r + 1$ and $d = 3$. Hence only $\mathcal{H}_2$ is MDS and $\mathcal{H}_r$ is not MDS for all $r > 2$. [3]

(c) i $\forall c(x) \in \mathcal{C}$, write $c(x) = u(x)g(x) + r(x)$ where $\deg(r(x)) < \deg(g(x))$.

As cyclic codes are linear, $r(x) = c(x) - u(x)g(x) \in \mathcal{C}$. At the same time, $g(x) \in \mathcal{C}$ is of the least degree among all nonzero polynomials. It concludes $r(x) = 0$. Hence $c(x) = u(x)g(x)$. [2]

ii In order to construct a BCH code on $\mathbb{F}_2$ with $n = 31$ and $d \geq 7$, consider cyclotomic cosets of 2 modulo 31, including

$$\begin{aligned}
&\mathcal{C}_0 = \{0\}\,,\ \mathcal{C}_1 = \{1, 2, 4, 8, 16\} = \mathcal{C}_2 = \mathcal{C}_4 = \mathcal{C}_8 \\
&\mathcal{C}_3 = \{3, 6, 12, 24, 17\} = \mathcal{C}_6 \\
&\mathcal{C}_5 = \{5, 10, 20, 9, 18\} \\
&\mathcal{C}_7 = \{7, 14, 28, 25, 19\}\,.
\end{aligned}$$

The generator polynomial is given by [3]

$$\begin{aligned}
g(x) &= \mathrm{lcm}(M^{(1)}(x), \cdots, M^{(6)}(x)) \\
&= M^{(1)}(x)M^{(3)}(x)M^{(5)}(x),
\end{aligned}$$

where

$$M^i(x) = \prod_{j \in \mathcal{C}_i} (x - \alpha^j).$$

[2]

iii Let $\boldsymbol{c} \in \mathcal{C}$ and write $\boldsymbol{c} = [c_0, c_1, \cdots, c_{n-1}]$. Since $g(x)|c(x)$, $\alpha, \cdots, \alpha^{n-k}$ are roots of $c(x)$. Hence

$$\boldsymbol{c} \cdot \begin{bmatrix} 1 & \alpha & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{n-k} & \cdots & \alpha^{(n-k)(n-1)} \end{bmatrix}^T = \boldsymbol{c}\boldsymbol{A}^T = \boldsymbol{0}.$$

As every subset of $n-k$ columns of $\boldsymbol{A}$ are linearly independent, $d(\mathcal{C}) \geq n-k+1$. At the same time, by Singleton bound $d(\mathcal{C}) \leq n-k+1$. The only possibility is $d = n - k + 1$. [3]

In fact, this cyclic code is Reed-Solomon code.